Foundations of Generative AI (ICT4403) - Assignment 2

Department of Information & Communication Technology
Assignment Submission & GitHub Repository Link

Student Name	Aditanshu Sahu
Registration Number	230953444
Assignment Title	Foundations of Generative AI (ICT4403) - Assignment 2
Title of Developed System	Wonderland Word Generator: Comparative Autoregressive Text Generation using Simple RNN, Bidirectional RNN, LSTM, and GRU
Name and Link of Dataset	Alice's Adventures in Wonderland by Lewis Carroll https://www.gutenberg.org/ebooks/11
GitHub Repository Link	https://github.com/aditans/wonderland-word-generator

Executive Summary & Project Deliverables

This project implements an autoregressive text-generation system trained on Lewis Carroll's 'Alice's Adventures in Wonderland' obtained from Project Gutenberg. Four distinct recurrent architectures—Simple RNN, Bidirectional RNN, LSTM, and GRU—were systematically trained under strictly controlled experimental conditions (matching sequence length, embedding dimensions, recurrent units, optimization hyperparameters, and loss objectives).

All code, Jupyter notebook, trained checkpoints, loss/accuracy curves, perplexity comparisons, runtime benchmarks, and qualitative sampling temperature generations (T = 0.5, 0.8, 1.2) are fully documented and open-source in the GitHub repository linked above.

- Jupyter Notebook (.ipynb) & Modular Python Pipeline (.py) in repository root and notebooks/
- Autoregressive rolling sequence dataset construction with Project Gutenberg text cleaning
- Comparable implementation of Simple RNN, Bidirectional RNN, LSTM, and GRU
- Quantitative Evaluation: Loss curves, validation accuracy, perplexity, and training runtime
- Qualitative Evaluation: Seed generation across three sampling temperatures (T=0.5, 0.8, 1.2) assessing coherence, diversity, and repetition
- Comprehensive README.md with full analysis, benchmark tables, and best model justification